César Garin

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SUMMARY

EPFL Computer Science and Communication Systems student with a strong quantitative foundation in probability, statistics, machine learning, algorithms, and scientific computing. Builds independent projects across predictive modelling, NLP, reinforcement learning, and financial data analysis. Currently on a one-year exchange at DTU (Copenhagen), focusing on statistical learning and quantitative modelling. **Available for a 3-month internship, June–August 2027**; open to projects and part-time roles in 2026–27.

WORK EXPERIENCE

Summexa AI **Remote**

Founder & Developer – solo project, pre-launch prototype *Jul 2026 – Present*

- Built a **self-hosted AI business analyst** for SMEs: a local LLM (**Llama 3** via Ollama) with a **LangChain/ChromaDB RAG pipeline** over company files (PDF, Word, Excel, CSV), multilingual chat, and automated analysis and weekly reports.
- Engineered it as a **multi-tenant FastAPI product** (JWT/bcrypt auth, **Docker Compose** deployment with HTTPS and one-click updates, **Stripe** per-seat licensing); defined pricing and go-to-market.

Independent Entrepreneur **Lyon, France**

Event Services *Jun 2026 – Present*

- Independently operate a **service business** providing services for **weddings, professional events and private events.**
- Generated **€4,000+** in revenue since launching the activity.

Getronics **Lyon, France**

Intern *Jan 2024 – Mar 2024*

- Completed **cross-departmental** rotations within an international IT services company.
- Gained exposure to corporate finance, revenue mechanisms, budgeting, cost analysis, and procurement.

EDUCATION

EPFL — École Polytechnique Fédérale de Lausanne **Lausanne, Switzerland**

B.Sc. Computer Science & Communication Systems *Sep 2024 – Jun 2027*

- Coursework: Machine Learning, Algorithms, Probability & Statistics, Computer Architecture, Software Construction, Computer Systems **(TCP/IP networking)**, Linear Algebra and Mathematical Analysis.

DTU — Technical University of Denmark **Copenhagen, Denmark**

B.Sc. Computer Science – Academic Exchange Year *Sep 2026 – Jun 2027*

- Coursework: Machine Learning for Signal Processing, Algebra, Intelligent Systems, Digital Communication, Database Systems, Logic for Security, Algorithms for Massive Data Sets, Software Development.

SELECTED PROJECTS

ML × Quantitative Finance Research Series **2026**

Personal Research

- Developed a **four-project research series** applying machine learning to financial modelling, covering **XGBoost** equity price prediction, **NLP** sentiment analysis with **FinBERT**, **PPO reinforcement learning** for derivative hedging, and **neural networks** for implied volatility surface modelling.

NanoGPT — Transformer Language Model from Scratch **2026**

Personal Project

- Implemented a **GPT-style transformer architecture** from scratch in **PyTorch**, including **multi-head self-attention, residual connections, layer normalization, and feed-forward blocks.** Built and trained the model of approximately **10 million parameters** on the Tiny Shakespeare dataset.

Mini RISC-V Processor **Sep 2025 – Jan 2026**

EPFL Project

- Designed and simulated a **32-bit multicycle RV32I processor** in Verilog (FSM control, CSRs, interrupts, memory-mapped I/O); wrote a bare-metal Game of Life in **RISC-V Assembly**.

SKILLS & LANGUAGES

- **LLM Engineering: RAG** & vector search (LangChain, ChromaDB), local LLM deployment (Ollama), agent orchestration & tool calling, prompt design, LLM evaluation.
- **Operating Systems:** Linux, Windows, macOS.
- **Programming: Python**, C, Java, Scala, R, LaTeX.
- **Machine Learning: PyTorch**, scikit-learn, **XGBoost**, FinBERT, Reinforcement Learning (**PPO**).
- **Data & Scientific Computing:** NumPy, Pandas, MATLAB, Jupyter Notebook.
- **Tools: Git**, **Docker**, FastAPI, Excel, PowerPoint, Overleaf.
- **Languages:** French (native), English (**C2 – CEFR**), German (**Intermediate**). **Interests:** Competitive Skiing, Football, Weightlifting & Martial Arts, Electric guitar & Piano.
- **Activities:** Forum EPFL logistics staff (150+ companies, 2025), PolySI integration week supervisor (2025).